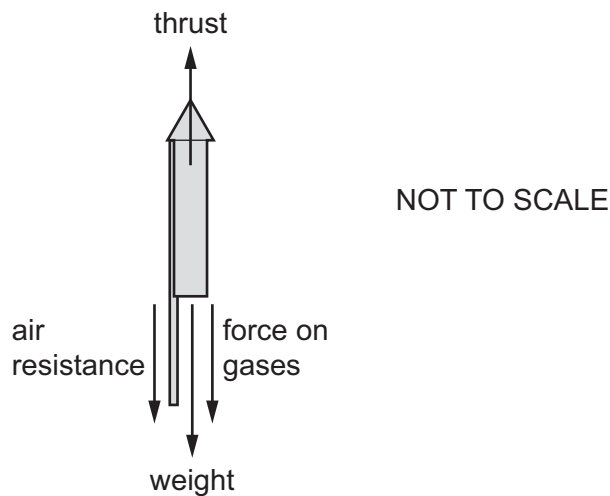


- 10 A firework travels vertically upwards in air. Gases are pushed vertically downwards from the firework.

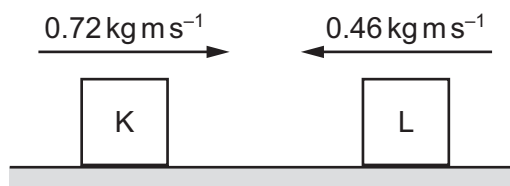
Several forces that act on the firework and gases are shown.



Which forces are a Newton's third law pair?

- A air resistance and force on gases
 - B air resistance and thrust
 - C force on gases and thrust
 - D thrust and weight
- 11 Two blocks K and L slide towards each other along a horizontal frictionless surface.

The diagram shows the momentum of the two blocks just before they collide.



During the collision, the blocks are in contact with each other for a time of 0.084 s .

After the collision, the blocks separate and block L moves back along its original path with a momentum of 0.12 kg m s^{-1} .

What is the magnitude of the average force exerted on block L by block K during the collision?

- A 3.1 N
- B 4.0 N
- C 6.9 N
- D 7.1 N